

Student Drop Out Risk Prediction Model

July 2026

- The dataset is taken from UCI Machine Learning Repository with the title "Predict Student's Dropout and Academic Success" based from the data of a high education institute in Portugal.
- There are **4424** entries of historical data with no missing values.
- The dataset comes with **36** features and a **target** that consists of three classes: Dropout, Enrolled, Graduate.
- The dataset is made of a mix of **categorical** and **numerical** features.

1 Data Cleaning:

- Fix typos, rename some features, and standardize the naming convention (**Example:** Nacionality becomes Nationality, Marital status becomes Marital_status).
- Check for duplicates in the dataset.
- Fix corrupted data and make sure numerical data is within boundary (**Example:** the maximum value of Curricular_units_1st_sem_grade is 20, so $1.73e + 16$ should be 17.3).

2 Feature Modifications:

- Convert data type for categorical features.
- Group some classes in features that have too many for a simplified classification.
- Map the target's classes as integers (Dropout: 0, Enrolled: 1, and Graduate: 2).

Exploratory Data Analysis (EDA)

1 Data Distribution Analysis:

- Calculated the percentage of the three Target classes (Dropout, Enrolled, Graduate) to show class imbalance.
- Checked the distribution percentage of the most common class of all **categorical features**.
- Checked the value dominance and the data distribution's skewness of all **numerical features**.

2 Feature Elimination:

• Categorical Features:

- Evaluate the feature with the majority class domination rate greater than 95% (Features with less than 95% domination rate are automatically kept)
- Eliminate a feature if the minority group has less than 30 sample size and the difference in dropout rate less than 15%.

• Numerical Features:

- Evaluate features with $|Skewness| > 4$.
- Isolate the minority slice of the data and calculate the dropout rate. If the minority slice has less than 30 count and the difference with the dropout rate from the baseline population is less than 15%, drop the feature.
- Replace some features that are dropped with the trajectory deltas (**Example:** Evaluations_Delta is the delta between the first and the second semesters' evaluations, Grade_Trajectory is the delta between the second and the first semesters' grade).

There are **27** remaining features that are not Target in the final dataset.

Correlation Matrix

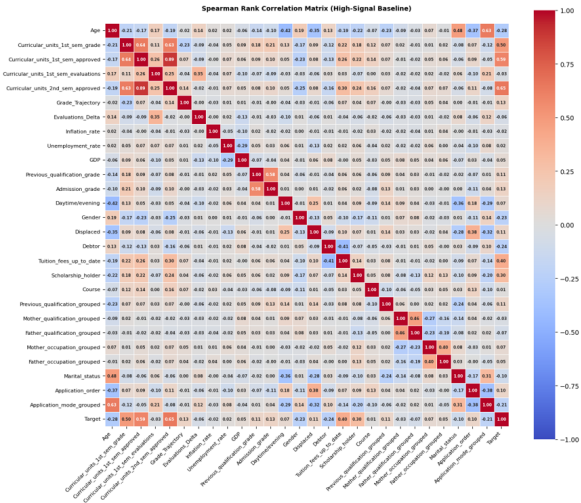


Figure 1: Spearman Rank Correlation Matrix for Final Dataset

Splitting Data

1 Multiclass Classification:

- **One-hot-encode** categorical features except for features that are already binary (**Example:** Gender).
- Set X as a list of all columns except for Target and y as the column Target.
- Split X and y as **train** and **test** data.
- Scale both train and test data with **StandardScaler**.
- Convert both train and test for y to **integer**.

2 Binary Classification:

- Merge the class in Target to be only two classes (1 and 2 merge as 0 which stands for safe from dropping out, while 0 relabeled as 1 which stands for drop out risk).
- Repeat the steps from **Multiclass Classification**.

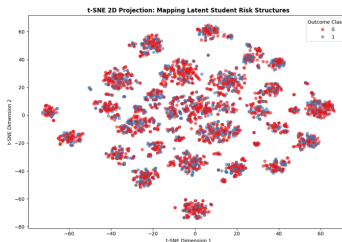
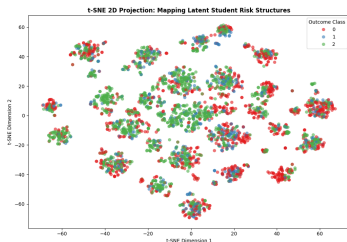
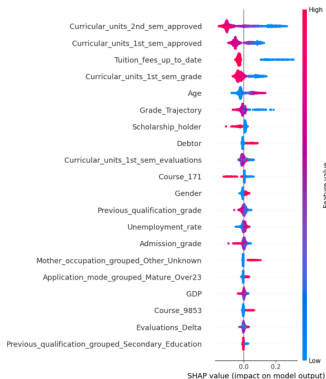


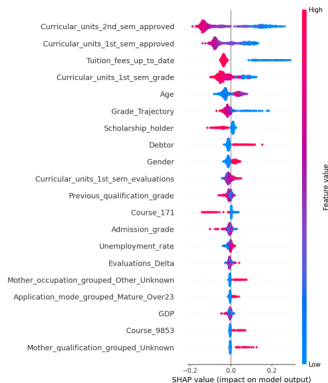
Figure 2: t-SNE Mapping for Multiclass and Binary Classification

Feature Importance

Train a random forest model and generate a **SHAP** (SHapley Additive exPlanations) summary plot.



(a) Multiclass Classification



(b) Binary Classification

The SHAP summary plot shows the top 20 most important feature for the decision making of the model with the top feature as the most important one.

Model

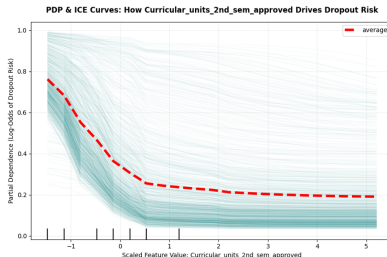
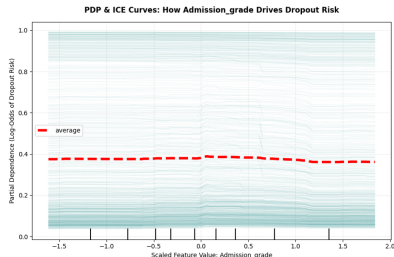
The models used are: **Random Forest**, **XGBoost**, **LightGBM**, **SVC**, and an **Ensemble Model** from the models mentioned. Use **RandomizedSearchCV** to do a 5-fold cross validation to search for the best parameters to tune the model.

Model	Accuracy	Precision (Dropout)	Recall (Dropout)	F1-Score (Dropout)
Random Forest Multiclassifier	0.76	0.79	0.75	0.77
XGBoost Multiclassifier	0.75	0.79	0.70	0.74
LightGBM Multiclassifier	0.75	0.77	0.71	0.74
SVC Multiclassifier	0.72	0.79	0.65	0.72
Ensemble Multiclassifier	0.75	0.78	0.71	0.75
Binary Random Forest	0.88	0.87	0.73	0.79
Binary XGBoost	0.87	0.79	0.82	0.80
Binary LightGBM	0.87	0.79	0.81	0.80
Binary SVC	0.87	0.79	0.80	0.79
Binary Ensemble	0.88	0.82	0.79	0.80
Tuned Binary Random Forest	0.87	0.80	0.78	0.79
Tuned Binary XGBoost	0.86	0.77	0.82	0.79
Tuned Binary LightGBM	0.87	0.78	0.83	0.80
Tuned Binary SVC	0.87	0.78	0.81	0.80
Tuned Binary Ensemble	0.88	0.81	0.81	0.81

The chosen model is Tuned Binary Ensemble that has the largest F1-Score. The model's drop out risk F1-Score gap between training and testing is only 1.68% which indicates that the model doesn't overfit.

Model Decision Making

This is a comparison of Partial Dependence Plots (PDP) and Individual Conditional Expectation (ICE) plots for the feature `Admission_grade` and `Curricular_units_2nd_sem_approved` using the Tuned Binary Ensemble model.



`Admission_grade` gives a flat curve meaning that it has no marginal influence to the model's predictions. In comparison, the second semester's approved units has massive impact to the model's predictions, as the lower the units approved the more the drop out risk.

Bias and Fairness Analysis

An analysis of outputs across sensitive groups Gender, Age Group (new feature with classes Traditional (≤ 21), Young Adult (22-30), and Mature-Age (> 30)), and Debtor.

Sensitive Group	Demographic Parity Difference	Equalized Odds Difference	Disparate Impact Ratio
Gender	0.0323	0.0448	0.9057
Age Group	0.0341	0.0587	0.9022
Debtor	0.0273	0.1004	0.9161

- All sensitive groups have less than 5% demographic parity difference is achieved, which means that the different groups have a similar selection rate.
- The equalized odds difference shows that the model outputs for Gender and Age Group are similar with less than 10% difference. Debtor status tightly adheres to the standard compliance threshold at 10.04% due to ground-truth financial correlations.
- The disparate impact ratio fits within the industry standard of 80% to 125%, meaning the ratio of the selection rate between vulnerable group to the privileged group are fairly equal.

From this analysis it is concluded that there are no further need for mitigation as the model satisfies the industry standard.